

Realized-Volatility Forecasting and Volatility-Managed Sizing on the S&P 500

A Pre-Registered, Walk-Forward,
Multiple-Testing-Corrected Study — with an Honest Null

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Abstract

We study realized-volatility forecasting and volatility-managed exposure sizing on 21 years of one-minute S&P 500 data, conditioned on the VIX term structure, under a validation protocol fixed *before the first run*: pre-registered hypotheses, grids, costs and ranking rule; walk-forward parameter selection on training windows only; and multiple-testing corrections across the 73 configurations actually computed. Two results emerge, of opposite sign. *Forecasting holds up*: log-HAR attains a lower out-of-sample QLIKE than the random walk, and a nested mean-squared-prediction-error Clark-West test rejects equal accuracy ($p \approx 0.03$); the edge grows with horizon (MSE-based R_{OOS}^2 from +23 % at $h=1$ to +43 % at $h=22$), implied volatility adds information (HAR-IV, $p = 0.019$), and a rich linear feature set edges HAR-IV through *features* rather than regularization or nonlinearity. A loss-matched Gamma-GLM with the VIX term-structure slope is the strongest of a few candidates to lower *both* QLIKE (−4 %) and MSE, in both sub-periods (Diebold-Mariano $p = 0.0008$, exploratory). A lag diagnostic shows the forecast is well calibrated on average (Mincer-Zarnowitz slope ≈ 1 , top-forecast-decile ratio 1.08, tail coverage 9.8 %); its visible “lag” is largely mechanical — of the $2.54\times$ spike-underestimation a perfectly-calibrated forecaster already produces $\approx 1.90\times$, and only the residual (+0.64, $P < 0.001$) is genuine under-reaction to unanticipated shocks. *The trading strategy does not hold up*: the best walk-forward volatility-managed / VIX-term-structure combination (Sharpe 0.96, maximum drawdown

*Independent researcher (contact via the repository issue tracker). Code, the pre-registration protocol and the full result tables are provided in the accompanying repository; licensed raw data are not redistributed (sha256 fingerprints published). Research and educational project, not investment advice. All results are out-of-sample and, unless stated otherwise, net of assumed transaction costs on a price index. All numbers reproduce deterministically from the versioned code (seed 42) and the author bears full substantive responsibility.

-9.9% versus buy-and-hold $0.83 / -34.0\%$) is *disqualified by the study's own pre-registered rules* — probability of backtest overfitting 0.84 and a single-market effect — its Sharpe advantage lies inside the bootstrap noise band, and its clean draw-down profile is largely a volatility-level artifact. Every failure is reported. The contribution is the forecasting result and the validation methodology, not a tradeable edge; no result reaches **supported** status without a live prediction log.

Keywords: realized volatility, HAR, volatility-managed portfolios, VIX term structure, walk-forward analysis, backtest overfitting, deflated Sharpe ratio, null result.

JEL: C22, C53, C58, G11, G17.

I. Introduction

Two empirical regularities motivate the study. First, *volatility is forecastable and returns are barely so*: realized variance has strong, slowly-decaying autocorrelation, whereas daily index returns are close to unpredictable.¹ Second, *the reward-to-risk of equity exposure is not constant*, so scaling exposure inversely to expected volatility can raise risk-adjusted performance² — though this result is fragile out of sample.³

This paper is deliberately a methodology paper with an honest outcome. Its value is not a headline Sharpe but the discipline applied to a practical question, and the willingness to report — and *act on* — a null. The design rests on four commitments. (i) *Pre-registration*: hypotheses (with null and expected effect), the complete grids, the cost model, the out-of-sample split, and the ranking and disqualification rules were fixed in writing before the first campaign backtest. (ii) *Walk-forward everywhere*: no reported strategy uses parameters chosen on the data it is evaluated on; parameters are re-selected annually on trailing windows. (iii) *Multiple-testing accounting*: every configuration computed ($n = 73$) enters the deflated Sharpe correction, and the combination universe is additionally audited with the probability of backtest overfitting via combinatorially symmetric cross-validation.⁴ (iv) *Full reporting, including disqualification*: when the study’s own rules disqualify its best strategy, that becomes the result.

Scope. We publish the S&P complex only (SPX + VIX) — the data whose provenance we fully stand behind. This choice interacts with the pre-registration in a way that is central to the paper: the fixed rules *disqualify single-market-only effects*, so a single-market strategy result *cannot* be a tradeable claim by the study’s own ex-ante standard. That is not a limitation worked around; it is the honest conclusion. The replication literature warns repeatedly how easily backtest “edges” arise from multiple testing and selective reporting,⁵ and the present protocol is built to disqualify its own favourite result

¹Andersen, Bollerslev, Diebold & Labys (2003); Corsi (2009).

²Moreira & Muir (2017).

³Cederburg, O’Doherty, Wang & Yan (2020).

⁴Bailey & López de Prado (2014); Bailey, Borwein, López de Prado & Zhu (2017).

⁵White (2000); Harvey, Liu & Zhu (2016); Bailey, Borwein, López de Prado & Zhu (2014).

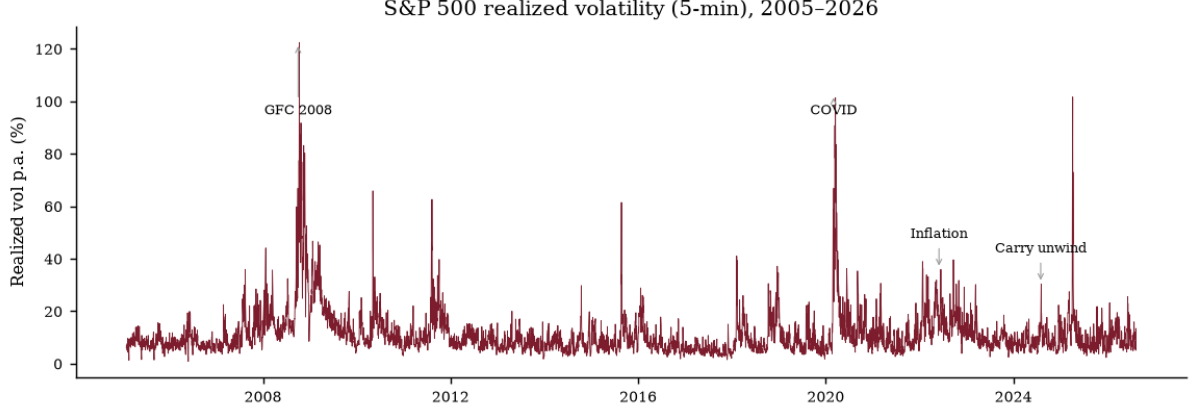


Figure 1: Annualized realized volatility of the S&P 500 from five-minute returns, 2005–2026. Volatility clusters and mean-reverts slowly — the raw material of the forecasting result.

rather than to survive review.

The remainder is organized as follows. Section II describes the data and realized measures; Section III the forecasting layer; Section IV the lag diagnostics that decompose the apparent forecast “lag”; Section V the strategy design and validation protocol; Section VI the disqualified campaign; Section VII implementation and costs; Section VIII a deliberately sharp self-assessment; and Section IX concludes.

II. Data and Realized Measures

Intraday. Licensed one-minute index candles (MarketTick) for the S&P 500, February 2005 to August 2026, regular trading hours (index prints, not exchange trades — the S&P 500 cash index is not itself tradeable); 5,543 daily session files, of which 5,417 enter the modelling panel after HAR warm-up and VIX alignment. The raw feed also carries roughly one hundred weekend-dated vendor files, all excluded by the trading-day and VIX alignment — the panel contains zero weekend dates; the out-of-sample window holds 3,422 trading days. Daily realized measures are built from five-minute sampling (realized variance, bipower variation, jumps, semivariances, realized quarticity),⁶ which is the standard microstructure-noise/efficiency compromise.⁷

⁶Andersen, Bollerslev, Diebold & Labys (2003); Barndorff-Nielsen & Shephard (2004).

⁷Liu, Patton & Sheppard (2015).

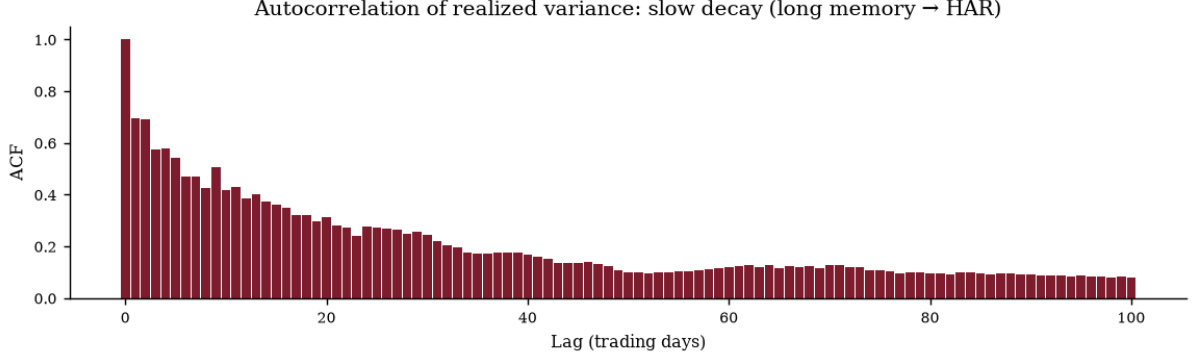


Figure 2: The autocorrelation of daily realized variance decays roughly hyperbolically — the long-memory signature the HAR model captures.

Volatility term structure. Two distinct sources. The *VIX level* used by HAR-IV ($\text{ivar} = (\text{VIX}/100)^2/252$) is daily-sampled from the **licensed MarketTick one-minute VIX feed** (`data/raw/vix`, a separate licensed feed of $\sim 5,280$ daily files), *not* FRED; only the term-structure slope $\ln(\text{VIX3M}/\text{VIX})$ is from FRED (`VXVCLS`, `VIXCLS`). Both are re-indexed to trading days, forward-filled at most five days, and lagged one day wherever used; reproducing the ladder from HAR-IV downward requires the MarketTick VIX feed, not FRED alone.

Conventions. (i) The S&P 500 is used as a price index: long-only CAGRs are understated by roughly the dividend yield; all candidates share the bias. (ii) The out-of-sample period runs from 2 January 2013; 2008 lies in the training region only. (iii) Weights for day t use information to the close of $t - 1$, enforced by lagging every signal; an automated look-ahead test guards the pipeline. (iv) The risk-free rate is set to zero and declared explicitly: Sharpe ratios are excess over zero and uninvested capital is unremunerated, applied uniformly to every sleeve and to buy-and-hold. (v) The cost model charges one basis point per unit of turnover, stress-tested at two and five basis points; overnight positions pay $2 \text{ bp} \times |w|$ per day; leverage is capped at three.

Table 1. Forecast ladder — S&P 500, $h = 1$, out-of-sample 2013–2026

QLIKE is the primary loss (lower is better); the R^2 column is the MSE-based out-of-sample R^2 relative to the random walk. Dashes mark values not computed for that model. All models run through the same expanding-window harness.

Model	QLIKE ↓	R^2_{OOS} (MSE)
Random walk (RV)	0.308	0 %
EWMA ($\lambda = 0.94$)	0.351	−20 %
HAR	0.247	+20 %
log-HAR	0.225	+23 %
HAR-IV (+VIX)	0.204	+30 %
OLS / Ridge (rich features)	0.197	—
Lasso / ElasticNet	0.198	—
Gradient boosting	0.205	—

III. The Forecasting Layer (the result that holds)

We evaluate a model ladder in an expanding-window out-of-sample harness (first forecast 2 January 2013, 3,422 days) under QLIKE loss, which is robust to noisy volatility proxies,⁸ with Diebold-Mariano tests⁹ and, for nested comparisons, Clark-West.¹⁰ The Clark-West statistic is computed on squared error (MSPE), the loss for which it is derived; QLIKE is reported separately as the ranking loss. The R^2 column of Table 1 is the familiar MSE-based out-of-sample R^2 ($1 - \sum \text{SE}_{\text{model}} / \sum \text{SE}_{\text{RW}}$), which need not track the QLIKE ranking.

Six findings follow, all from the tracked result files.

(a) *log-HAR beats the random walk* — QLIKE 0.225 versus 0.308, with a nested-MSPE Clark-West $p \approx 0.03$, the pre-registered confirmatory result.¹¹ The p -value is modest relative to the loss gap because the loss differences are heavy-tailed: a handful of spike days dominate their variance, so the mean reduction is large but the t -statistic is only about 1.9. The edge grows with horizon (MSE-based R^2_{OOS} +23 % at $h=1$, +27 % at $h=5$, +43 % at $h=22$).

⁸Patton (2011).

⁹Diebold & Mariano (1995).

¹⁰Clark & West (2007).

¹¹Pre-registered in PREREGISTRATION-FORECAST-EN.md (note QR002, frozen 2026-08-15): the single confirmatory hypothesis is $\text{QLIKE}(\text{log-HAR}) < \text{QLIKE}(\text{RW})$ on SPX / $h=1$ with Clark-West $p < 0.05$. Everything else in §III is exploratory.

(b) *Implied volatility carries incremental information.* HAR-IV improves on log-HAR (Clark-West $p = 0.019$), consistent in both sub-periods.

(c) *The linear feature set does the work, not regularization or nonlinearity.* Unregularized OLS reaches QLIKE 0.1968, matching Ridge (0.1968); Lasso and ElasticNet (0.1977/0.1978) are a statistical tie — we do not test these fourth-decimal differences (no Model Confidence Set here; see Section IX), so we read the linear family as a single plateau. All beat HAR-IV (Diebold-Mariano $p \approx 0.0003$ – 0.0008), while gradient boosting adds nothing (0.2045, $p = 0.86$). The honest reading: a rich realized-feature set helps slightly; the machinery does not. The QLIKE win does not carry to MSE, and unevenly so: the MSE-based R^2 is -9.7% for OLS and -11.4% for Ridge but -65% for Lasso/ElasticNet, while gradient boosting (which adds nothing on QLIKE) has the *best* MSE- R^2 of the ladder ($+33.5\%$). We therefore treat the linear-feature edge as fragile and do not build the sizing rule on it.

(d) *Daily-return GARCH cannot match the random walk* once intraday RV is available (QLIKE 0.456),¹² confirming that intraday information dominates daily-return conditional-variance models.

(e) *A multiplicity-aware check (Model Confidence Set).* To hold the forecasting layer to the same standard as the strategy, we compute a Model Confidence Set¹³ (QLIKE loss, stationary block bootstrap, 1,000 replications, block 10, seed 42) over the ten classical ladder models {RW, EWMA, AR(1), HAR, HAR-CJ, HAR-RS, HARQ, LHAR, log-HAR, HAR-IV}. The 90 % MCS is {HAR-IV} alone — every simpler model is excluded at $p < 0.01$, so HAR-IV is the unambiguous *classical* winner. Admitting the rich-feature/ML models widens the set: the 90 % MCS *appears* to widen to {Lasso, ElasticNet, HAR-CJ} with **HAR-IV excluded at $p = 0.013$** — but this second run is computed on a *different* loss matrix (the unfiltered `loss_matrix_classical`, on which the same model names carry different QLIKE: HAR-CJ 0.564 vs. 0.256, LHAR 1.852 vs. 0.684), so the two MCS statements are not directly comparable. That run is moreover *degenerate*: HAR-CJ (QLIKE 0.564, second-worst) sits inside the set while HAR-IV (0.204, third-best) is

¹²Bollerslev (1986); the textbook result is due to Andersen, Bollerslev, Diebold & Labys (2003).

¹³Hansen, Lunde & Nason (2011).

outside — an unstable MCS on this matrix, not a genuine displacement (on the unfiltered matrix without the ML columns the 90 % set is simply {HAR-IV}). We therefore do not read it as the feature family displacing HAR-IV, and do not build the sizing rule on it, both because that QLIKE win rides the -65% MSE- R^2 of Lasso (c) and because the comparison is not apples-to-apples.

(f) *The one genuinely new candidate: a QLIKE-native GLM with the VIX term-structure slope.* Adding the log VIX3M/VIX slope to HAR-IV and fitting it with a Gamma-GLM (log link) — that is, estimating *in the loss the model is judged by* rather than least squares on log-RV followed by a retransform — reaches QLIKE 0.1959 (-4% versus HAR-IV) while *also* improving MSE (its MSE- R^2 is $+33\%$, unlike the Lasso feature winner at -65% ; c), with Diebold-Mariano $p = 0.0008$ and a lower QLIKE in both sub-periods. It clears the same acceptance gates as the confirmatory result — but as a *post-hoc, exploratory* specification (developed after observing the QLIKE/MSE mismatch, and judged on Diebold-Mariano rather than the pre-registered Clark-West), so its p -value is not a nominal confirmatory one. Under the pre-registered Clark-West test it does *not* clear the bar: CW gives $p \approx 0.027$ against the HAR-IV baseline (≈ 0.037 against the correct nesting parent), both above the Bonferroni-adjusted threshold $0.05/6 \approx 0.0083$ the confirmatory arm uses; the $p = 0.0008$ figure comes from Diebold-Mariano’s different reference distribution, not null-calibrated for a (near-)nested pair. It is the concrete payoff of coupling the estimator to the evaluation loss — the fix for the fragility flagged in (c) — and remains a candidate, not **supported**: like everything here it awaits a live prediction log. This, the IV term-structure slope under a loss-matched estimator, is the study’s only forecasting result that even points beyond replication.¹⁴

For the sizing rule we deliberately use log-HAR, not the best model: it is the simplest member of the winning family, needs no auxiliary series at forecast time, and avoids stacking a second selection layer on the strategy.

¹⁴The natural comparator here is Bollerslev, Patton & Quaedvlieg (2016), whose HARQ uses realized quarticity to the same end; a like-for-like comparison is left to future work.

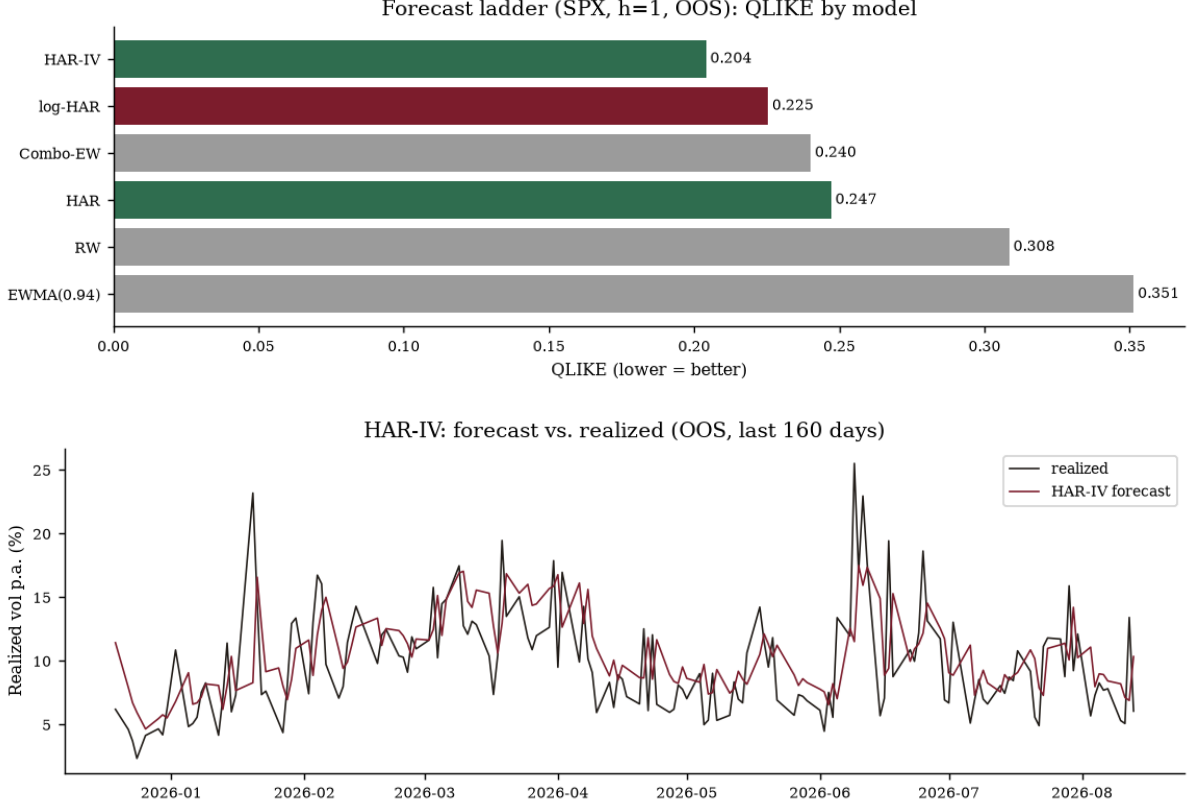


Figure 3: Top: QLIKE by model (the level-target LHAR/HARQ/HAR-CJ variants are fit artifacts and are omitted). Bottom: HAR-IV one-day-ahead forecasts tracking realized volatility.

IV. Lag Diagnostics: What “the Forecast Lags” Really Means

A recurring visual impression — that the forecast “lags” realized volatility — resolves into two very different things, only one of which is a defect. Table 2 measures forecast quality beyond the mean loss.

On the average day the forecast does not lag in any harmful sense. The Mincer-Zarnowitz slope is ≈ 1 (HAR-IV 1.016; the joint $a = 0$, $b = 1$ unbiasedness test is not rejected, HAC F $p = 0.53$) and the error autocorrelation is ≈ 0 (-0.02): no systematic, exploitable lag-1 trailing error. Much of what looks “smoothed/late” on a chart is correct behaviour — the forecast deliberately does not chase daily noise (five-minute RV is itself a noisy proxy of latent volatility). The random walk, by contrast, is visibly damped (slope 0.70) with strongly negatively autocorrelated errors (-0.33): it over-reacts and reverts.

Table 2. Forecast quality beyond average loss — out-of-sample HAR family

MZ slope: Mincer-Zarnowitz regression realized $\sim a + b \cdot \text{forecast}$ (1 = calibrated slope). The joint $a = 0$, $b = 1$ unbiasedness test is not rejected for HAR-IV (HAC F $p = 0.53$). Error

AC(1): lag-1 autocorrelation of log errors (≈ 0 = no exploitable linear lag-1 structure).

$\text{corr}(t) / \text{corr}(t-1)$: correlation of the forecast with same-day versus previous-day realized volatility. $\text{spike}\times$: mean realized / forecast on the top-decile up-jump days — these days are selected on the realized outcome, so a ratio > 1 is partly mechanical (regression to the mean under a right-skewed conditional law); the informative content is the ordering across models under a common selection, not the absolute level.

Model	MZ slope	err	AC(1)	corr(t)	corr($t-1$)	spike $\times$
Random walk	0.70	−0.329	0.779	1.000	3.98	
log-HAR	1.07	−0.008	0.800	0.964	2.73	
HAR-IV	1.02	−0.021	0.815	0.961	2.54	
C2 (rich + IV-slope)	0.83	−0.029	0.828	0.946	2.54	

The phase is “yesterday-heavy,” but mostly intrinsically. HAR-IV correlates 0.96 with yesterday’s realized volatility and 0.82 with the day it forecasts — the visible lag. This is largely unavoidable: realized volatility is highly persistent, so forecasting a slow process is forecasting something close to yesterday. Leading information moves the phase measurably forward — the rich + slope model has the highest same-day correlation (0.83) and the least dependence on yesterday (0.95) — but only at the margin.

The expensive lag is the spikes — and we identify how much of it is real. On the worst 10% up-jump days realized volatility averages $2.5\times$ the forecast (HAR-IV $2.54\times$; the random walk $3.98\times$). This is the bulge the eye sees — but the absolute ratio overstates the defect, because conditioning on the realized jump forces an excess above one even for a perfect forecaster. We separate the two. *The mechanical floor:* treating the HAR-IV forecast as the true conditional mean and drawing realized RV from the implied lognormal, a perfect forecaster mechanically produces $\text{spike}\times = 1.90$ [95% 1.80, 2.01] — about three-quarters of the observed 2.54. *A genuine residual:* the observed 2.54 lies above that floor with $P < 0.001$ (excess +0.64): there is real under-reaction to shocks the model cannot anticipate. *Otherwise the forecast is well calibrated:* binned by forecast decile (not outcome), mean-realized / mean-forecast is ≈ 1.0 in every bin including the highest (1.08), and realized exceeds the 90th predictive percentile on 9.8% of days ($\approx$ the nominal 10%). The reading: on the days the model itself calls high-vol it is right; the

Lag diagnostics: the forecast is well-calibrated on average; the residual 'lag' is the spike underestimation

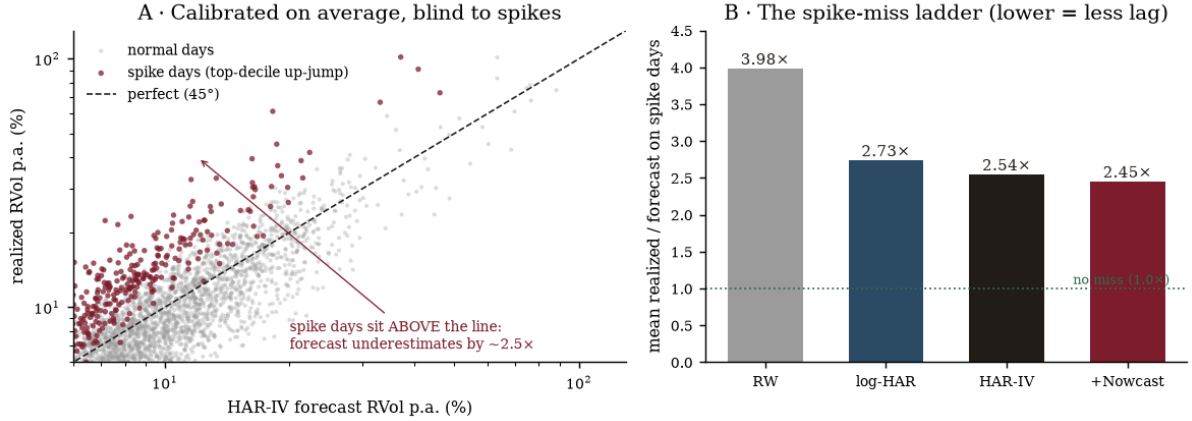


Figure 4: (A) HAR-IV forecast versus realized RVol: well calibrated across the bulk, but the top-decile up-jump days (red) sit above the 45° line — systematically underestimated. (B) The spike-miss shrinks with better information (features, implied vol, an intraday nowcast) but not to one.

residual miss is confined to the outcome-selected shock days, where features and implied vol shrink it (RW 3.98 $\rightarrow$ log-HAR 2.73 $\rightarrow$ HAR-IV 2.54) but a floor remains.

Can the spikes be nowcast away? We test the two same-day channels the data allow, incrementally on HAR-IV (nested Clark-West), moving the forecast origin from the previous close to the morning of day t (Table 3). Two honest findings. First, the overnight gap is nearly useless day-ahead and only a small, non-significant help same-day (-1.3% , $p \approx 0.18$), and it does not move the spike miss (2.54 $\rightarrow$ 2.49): the largest RTH spikes are not gap-driven but form inside the session. Second, intraday “RV-so-far” is the strongest single lever we found — the first hour lowers QLIKE by 4.8% ($p = 0.089$), roughly three times the overnight nowcast — yet it still only trims the spike miss to 2.45 $\times$, because on the spike days the first hour is only mildly elevated (median morning-RV percentile 0.59). The worst spikes are realized after the first hour, so not even a same-day nowcast sees them coming. None of these channels clears the multiple-testing bar; they are reported as the honest ceiling of what these data support.

This is *consistent with* the strategy result rather than a proof of it. A volatility forecast used to size risk must be right precisely on the days that matter — the spikes — and there the miss is largest and does not vanish with more history, the overnight gap, or a morning nowcast; a risk measure that under-reacts on the worst day de-risks a day late, after the

Table 3. Same-day information versus day-ahead (incremental on HAR-IV)

“RV-so-far” is the realized variance of the first 30/60 one-minute bars of day t , known by 10:00/10:30; the target is the full-session five-minute RV of the same day. All leakage-free.

None clears the Bonferroni threshold ($k = 5$ tested, $0.05/5 = 0.010$). CW p : nested Clark-West.

Added predictor (origin)	QLIKE	Δ QLIKE	CW p	spike $\times$
HAR-IV baseline (prev. close)	0.2041	—	—	2.54
+ overnight gap (day-ahead)	0.2037	−0.2 %	0.10	2.52
+ overnight gap (same-day)	0.2015	−1.3 %	0.18	2.49
+ RV-so-far 30 min (nowcast)	0.1964	−3.7 %	0.11	2.47
+ RV-so-far 60 min (nowcast)	0.1942	−4.8 %	0.09	2.45
+ RV-so-far 60 min + gap	0.1940	−5.0 %	0.07	2.43

drawdown is under way. The visible “lag” and the null trading result plausibly share this mechanism — but the trading null already has three independent, better-identified causes (Sections VI–VII), so we offer the spike-miss as a coherent narrative, not a fourth proof. The *target* matters too: RTH-only RV discards the overnight component — a median 6.8 % but a mean 15.7 % of total daily variance, exceeding 30 % on a fifth of days — so for a genuine total-risk objective the defensible fix is to move overnight into the target ($\text{RV}_{\text{total}} = \text{gap}^2 + \text{RTH-RV}$) rather than to use it as a feature.

V. Strategy Design and Validation Protocol

S1 — Volatility-managed exposure (VMG). $w(t) = \min(\tau(t-1)/\hat{\sigma}(t), c)$, where τ is the trailing 63-day mean realized volatility, $\hat{\sigma}$ the log-HAR one-day-ahead volatility forecast (information $\leq t-1$), and c the leverage cap. Net return $w \cdot r - 1\text{bp} \cdot |\Delta w|$. High-volatility days do not carry proportionally higher expected returns,¹⁵ so scaling down ahead of turbulence harvests the premium with fewer of the worst days. Walk-forward re-selects (target window, cap) annually from $\{21, 42, 63, 126\} \times \{1.5, 2, 3\}$.

S2 — VIX term-structure conditioning (VIXTS). Exposure keyed to VXV/VIX: full in contango (the volatility risk premium),¹⁶ reduced in backwardation. Unlike S1 the signal is implied, so it complements the realized channel. Supporting sleeves are TSMOM

¹⁵Moreira & Muir (2017).

¹⁶Whaley (2009); Simon & Campasano (2014).

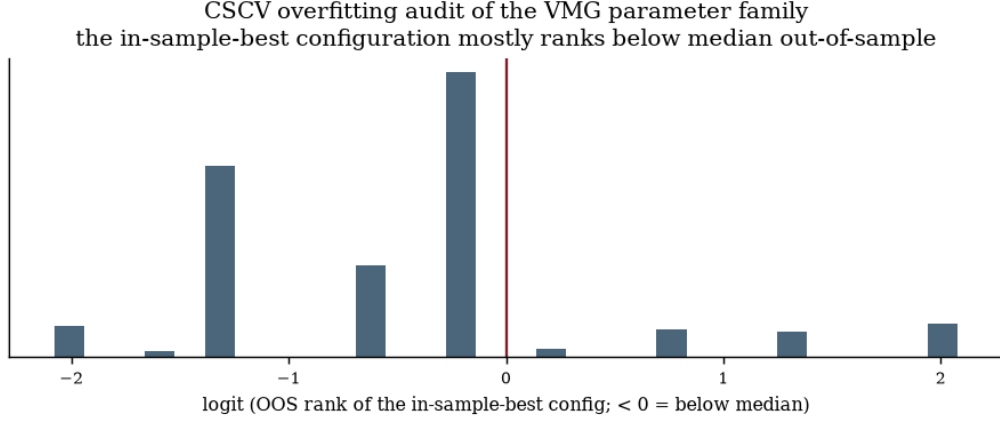


Figure 5: CSCV overfitting audit (illustrative $2 \times 2 \times 2$ VMG grid). In the full campaign the VMG parameter family has $\text{PBO} = 0.79$ (HAR-IV; 0.75 for log-HAR) and the combination universe $\text{PBO} = 0.84$ — both above the 0.7 disqualification gate; the in-sample-best configuration mostly ranks below median out-of-sample.

(long/flat, vol-targeted; the short leg is rejected decisively) and short-term reversal. A single-market combination weights these sleeve return streams inversely to their trailing 63-day volatility.

Protocol. (1) *Pre-registration*¹⁷: six hypothesis families with null and expected effect; the complete grids; costs (1/2/5 bp); the out-of-sample split; the composite ranking score $0.30 \cdot \text{Sortino} + 0.25 \cdot \text{Calmar} + 0.20 \cdot \text{Sharpe} + 0.15 \cdot \text{YearConsistency} + 0.10 \cdot \min(\text{PF} - 1, 1) \cdot 5$; and — decisive here — the disqualification rules: $\text{DSR} < 0.5$, $\text{PBO} > 0.7$, single-market-only effect, or a synthetic-data-only basis. Nothing was added after the first result. (2) *Walk-forward*: anchored five-year train, one-year test, stepped annually; only concatenated test segments are ranked or combined. (3) *Multiple testing*: deflated Sharpe ratio¹⁸ against the expected maximum of $n = 73$ SPX-only trials; PBO via combinatorially symmetric cross-validation;¹⁹ block-bootstrap Sharpe intervals (21-day blocks). (4) *Distributional risk*: fixed 21-day block-bootstrap Monte-Carlo (5,000 paths, seed 42).

¹⁷The campaign protocol is `PREREGISTRATION-EN.md` (QR006); the headline forecasting claim is pre-registered separately in `PREREGISTRATION-FORECAST-EN.md` (QR002). Both are released.

¹⁸Bailey & López de Prado (2014).

¹⁹Bailey, Borwein, López de Prado & Zhu (2017).

Table 4. The campaign combination (net, out-of-sample, SPX)

The best walk-forward volatility-managed / VIX-term-structure combination against single-sleeve VMG and buy-and-hold. Sharpe intervals are 21-day block-bootstrap 95 % intervals. P(drawdown > 20 %) is from the fixed-block bootstrap Monte-Carlo (only the forecasting MCS uses a stationary bootstrap).

Metric	Best combo	VMG (single)	Buy & Hold
Sharpe [95 % CI]	0.96 [0.44, 1.46]	0.95	0.83 [0.28, 1.34]
Maximum drawdown	−9.9 %	−25.1 %	−34.0 %
Calmar	0.62	0.57	0.39
Deflated Sharpe ($n = 73$)	0.99	0.97	0.96
MC P(drawdown > 20 %)	3.5 %	72 %	91 %

VI. Results: the Disqualified Campaign

Walk-forward parameter selection resolves the well-known overfitting paradox of volatility management at the family level: the fixed in-sample-best VMG parametrization is overfit (PBO = 0.79 for HAR-IV, 0.75 for log-HAR on SPX; `thesis_validation.csv` — though the same file reports PBO = 0.365 on NDX, so this overfitting flag is market-specific, a caveat the SPX-only headline should carry),²⁰ yet the family, honestly re-selected, delivers a positive walk-forward Sharpe. The combination then looks good on paper (Table 4).

But the campaign fails its own pre-registered gates. *Single-market effect* → *disqualified*: by construction every SPX-only combination is a single-market result, and the fixed rules disqualify single-market-only effects. *PBO = 0.84* → *disqualified*: the probability that the selection is overfit is 0.84, far above the 0.7 gate. *No significant Sharpe advantage*: Δ Sharpe over buy-and-hold is +0.13, but the 95 % Sharpe intervals overlap heavily ([0.44, 1.46] versus [0.28, 1.34]). *The clean tail is mostly a volatility artifact*: the combination runs at $\approx 6.5\%$ volatility versus $\approx 17\%$ for buy-and-hold; normalized to a common risk level the drawdown advantage shrinks toward parity, and single-market VMG alone matches or beats the combination on vol-matched Calmar. We therefore report no tradeable edge.

Cost stress and the graveyard. VMG and VIXTS survive five basis points — though VIXTS survives the *cost* stress, not its own hypothesis: the pre-registered Q015 (VIX-

²⁰Consistent with Cederburg, O’Doherty, Wang & Yan (2020).

term-structure sizing improving Calmar by +0.05 to +0.15) reaches only $\Delta\text{Calmar} \approx +0.023$ at its best grid point, with five of six grid points negative, so it misses its pre-registered band and is recorded *not supported*; and Q012 (TSMOM, correlation < 0.5 to VMG) measures 0.751, violating its own condition. The cost-fragile families die where they should — overnight carry collapses to Sharpe -2.38 at five basis points despite a real gross anomaly ($+7.3\%$ /yr overnight versus $+6.4\%$ intraday). Reported failures in full: long/short momentum (-52% SPX drawdown; the -68% sometimes quoted was the DAX sleeve in the earlier cross-market version), turn-of-month (a single-market artifact), net overnight carry, and a six-test mechanism battery in which only HAR-IV (implied volatility helps, $p = 0.019$) carried evidence; the other five were null or weak.

Why the trading result is null — and why that is the point. The volatility-managed idea is real (it survives walk-forward at the family level); the tradeable claim is not established, for three independent reasons that all point the same way: on a single market the effect is disqualified by the ex-ante rule; the selection among combinations is overfit (PBO 0.84); and the return-metric advantage is inside the bootstrap noise while the risk-metric advantage is largely a volatility level, not diversification. Any one would be enough. A fourth, mechanistic reading is consistent with them (though less cleanly identified, Section IV): the sizing forecast under-reacts most on exactly the spike days a risk overlay exists to catch, so it de-risks a day late. A paper that buried these behind a 0.96 Sharpe would be exactly the kind of result the replication literature warns about;²¹ reporting them is the contribution.

VII. Implementation and Costs

The backtest is on the S&P 500 price index, which is not directly tradeable; a live implementation would use ES/MES futures, whose all-in cost (bid/ask, roll, slippage in volatility spikes) exceeds the one basis point modeled here — the two/five basis-point stress is the honest bracket, and the overnight-carry collapse shows the strategies live

²¹White (2000); Harvey, Liu & Zhu (2016); Bailey, Borwein, López de Prado & Zhu (2014).

close to their cost breakeven. Signals are formed at or near the close (a fifteen-minute approximation versus the exact close used in the backtest, documented as a gap), executed market-on-close, with profit and loss aligned close-to-close. None of this rescues the null result; it is stated so the reader can see the execution assumptions rather than infer them.

VIII. Critical Assessment

(1) *Forecasting is mostly replication.* log-HAR $\succ$ RW and HAR-IV are known; the contribution there is a clean, leakage-tested, single-market replication with honest tests. The one genuine increment is the loss-matched Gamma-GLM with the VIX term-structure slope (Section III f) — it clears the DM gate but *not* the pre-registered Clark-West gate (the honest test for this near-nested pair), so it stays exploratory and unconfirmed absent a live log. (2) *The strategy is beta management at best*, and here not even that survives selection: no short book, no market-neutral source, and on one market the diversification argument is thin. (3) *Statistical honesty about ranks and differences:* the deflated Sharpe of the leaders is high only because restricting to SPX shrinks the trial count ($227 \rightarrow 73$); we report this openly and rest the conclusion on the PBO/single-market disqualification, not on the flattering DSR. (4) *The decisive scenarios are out of sample:* no 2008 inside the OOS window, no slow-grind bear anywhere in the data. (5) *No live track record:* a pre-registered, micro-size prediction log is the only route from **testing** to evidence; every number here deserves a 30–50 % haircut until it exists. (6) *The lag/spike diagnostic is where method meets substance:* decomposing the “forecast lag” into a calibrated mean and a residual spike-miss, and relating that spike-miss to why the sizing strategy struggles, is the paper’s freshest piece of reasoning — with the caveat that the spike ratio conditions on the outcome and so is a coherent narrative, not a clean identification. On our own honest ledger: methodology strong, execution solid, novelty deliberately modest.

IX. Conclusion

On 21 years of intraday S&P 500 data, evaluated under a pre-registered, walk-forward, multiple-testing-corrected protocol: (1) realized volatility is forecastable — log-HAR beats the random walk, implied volatility adds information, and the gain over HAR comes from features rather than regularization or nonlinearity; (2) the corresponding volatility-managed / VIX sizing strategy, however good it looks, is disqualified by the study’s own rules and is reported as a null. The failures are reported with equal weight. The publishable, reproducible contribution is the forecasting result plus the validation methodology — and the demonstration that a disciplined protocol will, and should, kill one’s favourite backtest.

Future work, in order of value: (1) a live prediction log, the only route from **testing** to **supported**, paired with a real ES/MES execution and option-chain cost model; (2) deciding the question up front — a forecasting paper or a strategy paper, the latter requiring multi-asset data from the outset because the single-market disqualification cannot be satisfied on the S&P complex alone; (3) coupling loss and estimator throughout, making the QLIKE-native Gamma-GLM the default fitter; (4) exploiting the one-minute data already in hand — noise-robust estimators, overnight variance as a target component, and the intraday nowcast track; (5) acquiring horizon-matched leading inputs (VIX9D, VVIX, SKEW); (6) extending the Model Confidence Set to the full feature/ML-inclusive ladder, adding rolling-origin cross-validation, block-bootstrap confidence intervals on loss differences, and a de-risk-before-drawdown hit-rate as a first-class outcome; and (7) modelling the distribution, not just the point, and coupling it to sizing.

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